Notes: Lazear & Rosen (JPE, 1981)
Rank-Order Tournaments as Optimum Labor Contracts

Contents

1 Big picture

- **Question.** Why do many labour markets and internal organizations use rank-based pay, prizes, promotions, and large wage spreads rather than paying workers directly according to measured output?
- **Core idea.** A rank-order tournament pays workers according to ordinal rank rather than cardinal output. The worker is rewarded for beating rivals, not for the exact distance by which output exceeds rivals' output.
- **Main theoretical result under risk neutrality.** A two-person tournament can induce the same efficient skill investment as a piece-rate contract. The correct prize spread makes the marginal benefit of improving the probability of winning equal to the marginal social value of productivity.
- **Why this is surprising.** The tournament is a nonlinear, discrete compensation scheme, while a piece rate is a linear transformation of output. Yet, with risk-neutral workers and competitive firms, both can implement the first-best investment level.
- **Why tournaments may dominate in practice.** Measuring rank can be cheaper than measuring absolute output. This is especially relevant for executives, professionals, academics, and other occupations where individual output is difficult to measure cardinally.
- **Risk-aversion result.** With risk-averse workers, the comparison between piece rates and tournaments is no longer trivial. Tournaments truncate some output risk but impose binomial prize risk. Either scheme can dominate depending on risk aversion, noise, and the shape of utility.
- **Common shocks result.** Rank-order pay can insure workers against shocks common to all contestants in the same activity or firm because common shocks drop out of the comparison.
- **Sorting result.** With heterogeneous worker ability and asymmetric information, tournament markets need not sort efficiently. Low-quality workers may try to enter high-quality leagues because high-quality contests have more attractive prizes.
- **Handicap result.** If worker types are known, competitive handicapping can restore efficient mixed contests by offsetting type differences while preserving incentives.
- **Economic takeaway.** Large promotion prizes and executive pay gaps need not reflect only contemporaneous marginal product. They can be incentive devices that induce effort and investment earlier in a career.

2 Incremental contribution of the paper

2.1 What the paper adds relative to marginal-product wage theory

Traditional competitive labour-market theory emphasizes wages equal to marginal product. Lazear and Rosen show that this is too narrow for incentive environments. A worker's wage can exceed or fall below current marginal product because the wage is part of a contest prize structure. The prize structure is chosen to influence pre-contest effort or skill investment.

Interpretation: A president's salary may be high not because the president's current one-day productivity is many times higher than the vice-president's, but because the promotion prize motivates many contestants before the promotion is awarded.

2.2 What the paper adds relative to piece-rate incentive theory

Earlier incentive models focused on piece rates or output-contingent pay. Lazear and Rosen introduce rank-order pay as a parallel institution. They show that ordinal information can be enough to provide efficient incentives when cardinal measurement is costly.

Interpretation: This is the paper's key conceptual move: incentives do not require measuring the level of output if workers can be compared reliably.

2.3 What the paper adds to agency theory

In standard agency models, workers post bonds or receive output-based pay to align incentives. Lazear and Rosen reinterpret tournament prizes as a bond-gamble mechanism. Each contestant effectively stakes a bond and competes for the pool of bonds. The incentive is to win the gamble.

Interpretation: This reframes promotion ladders and executive wage spreads as governance devices, not merely compensation anomalies.

2.4 What the paper adds to labour sorting theory

The paper shows that tournament incentives interact with worker heterogeneity. When worker quality is private information, low-quality workers have incentives to enter contests populated by high-quality workers. This creates adverse selection and potential breakdown of efficient sorting.

Interpretation: The paper therefore connects incentive contracts to assignment and sorting. Tournaments do not merely motivate effort; they also affect who wants to compete with whom.

2.5 Why the contribution is still foundational

The paper became the canonical model of tournament theory in labour economics and corporate governance. It explains promotion prizes, executive compensation, wage inequality inside firms, rank-based evaluation, promotion ladders, and competition among employees.

Economic message: Its incremental contribution is to show that wage inequality can be an efficient incentive mechanism even when workers are not paid their realized marginal products.

3 Model primitives and measurement

This paper is theoretical, so the "data" section is a model-primitives section. The object being measured is not an empirical variable but the incentive effect of different compensation systems.

3.1 Workers, output, and investment

A worker chooses a costly investment or effort level μ before output is realized. Output is

$$q_i = \mu_i + \varepsilon_i, \tag{1}$$

where ε_i is luck, persistent individual-specific uncertainty, or productivity noise.

- μ_i is controlled by the worker and raises expected output.
- ε_i is random and has mean zero.
- Investment costs are $C(\mu_i)$ with $C' > 0$ and $C'' > 0$.
- The product market value of one unit of output is V.

Interpretation: The model separates the incentive component of output from random luck. This is essential because rank-order pay rewards observed output rank even though rank partly reflects luck.

3.2 Firm technology and competition

Production is additively separable across workers, and firms are risk neutral. Competitive entry implies zero expected profits.

Interpretation: The zero-profit condition makes expected wages equal expected product. Therefore, differences between piece rates and tournaments come from incentives and risk allocation, not monopoly profits.

3.3 Piece-rate contract

A piece-rate contract pays the worker in proportion to realized output. In the risk-neutral benchmark, the worker receives

$$y_i = rq_i - C(\mu_i), \tag{2}$$

where r is the piece rate.

Interpretation: Piece rates use cardinal output information. They are natural when individual output is easy to measure accurately, such as sales or simple production tasks.

3.4 Tournament contract

A two-person tournament specifies two fixed prizes:

$$W_1 > W_2, \tag{3}$$

where W_1 is paid to the worker with the higher realized output and W_2 to the loser. The prize spread is

$$\Delta W = W_1 - W_2. \tag{4}$$

Interpretation: Tournaments use ordinal information. The firm needs only to determine who performed better, not the exact value of each worker's output.

3.5 Probability of winning

For worker j competing against worker k,

$$P_j = \Pr(q_j > q_k) = \Pr(\varepsilon_k - \varepsilon_j < \mu_j - \mu_k) = G(\mu_j - \mu_k), \tag{5}$$

where G is the distribution of the difference in noise terms and g is its density.

In a symmetric equilibrium, $\mu_j = \mu_k = \mu$, so

$$P_j = \frac{1}{2}, \qquad \frac{\partial P_j}{\partial \mu_j} = g(0). \tag{6}$$

Interpretation: The incentive effect of a tournament depends on how much extra investment raises the probability of winning. That marginal probability is $g(0)$ in the symmetric benchmark.

3.6 Why measurement costs matter

The paper emphasizes that rank measurement can be cheaper than output measurement. Ordinal comparisons require less information than cardinal productivity measurement.

Interpretation: The tournament is attractive when the firm can cheaply rank workers but cannot cheaply measure each worker's absolute marginal product. This is why the theory is naturally applied to managers and executives.

4 Models and theoretical specifications

4.1 Risk-neutral piece rate

A risk-neutral worker maximizes expected income:

$$\max_{\mu} \; r\mu - C(\mu). \tag{7}$$

The first-order condition is

$$r = C'(\mu). \tag{8}$$

The firm's expected profit from one worker is

$$(V - r)\mu. \tag{9}$$

Competition for workers implies

$$r = V. \tag{10}$$

Therefore the piece rate induces

$$V = C'(\mu). \tag{11}$$

Interpretation: The piece rate implements the first-best investment level: the private marginal return to investment equals the social marginal product.

4.2 Risk-neutral tournament

Worker j's expected payoff in a two-person tournament is

$$P_j W_1 + (1 - P_j) W_2 - C(\mu_j). \tag{12}$$

The worker chooses μ_j to satisfy

$$\frac{\partial P_j}{\partial \mu_j}(W_1 - W_2) = C'(\mu_j). \tag{13}$$

In a symmetric equilibrium,

$$g(0)\Delta W = C'(\mu). \tag{14}$$

The zero-profit condition is

$$\frac{W_1 + W_2}{2} = V\mu. \tag{15}$$

To implement the first-best $V = C'(\mu)$, the required spread is

$$\Delta W = \frac{V}{g(0)}. \tag{16}$$

Interpretation: The tournament implements the same efficient investment as a piece rate if the prize spread is chosen correctly. The level of the average prize handles participation and zero profit; the spread handles incentives.

4.3 Bond-gamble interpretation of the optimal prize structure

The equilibrium prizes can be written as

$$W_1 = V\mu + \frac{V}{2g(0)}, \qquad W_2 = V\mu - \frac{V}{2g(0)}. \tag{17}$$

Interpretation: Each contestant receives expected product plus or minus an implicit bond. The workers compete over a fair gamble. The possibility of winning the bond pool induces investment.

Economic message: This explains why the winning prize can exceed current marginal product. The prize is not only compensation for the winner's realized output; it is also an incentive for all contestants.

4.4 Comparison with fixed standards

A worker can also be compared with a fixed standard rather than a rival. A high prize is paid if output exceeds the standard, and a low prize otherwise.

Interpretation: A fixed-standard scheme can also implement efficient investment. This reinforces the paper's broader point: efficient incentives do not require a piece rate. They require a schedule that makes the marginal probability of a higher reward respond correctly to effort.

4.5 Risk-averse piece rate

With risk aversion, the piece-rate contract pays

$$y = I + rq - C(\mu) = I + r\mu + r\varepsilon - C(\mu). \tag{18}$$

The worker's first-order condition remains

$$r = C'(\mu), \tag{19}$$

because the noise term is independent of investment. The firm chooses I and r subject to zero profit.

The paper's marginal condition for the optimal piece rate is

$$[V - C'(\mu)]\,\frac{d\mu}{dr}\mathbb{E}[U'(y)] + \mathbb{E}[\varepsilon U'(y)] = 0. \tag{20}$$

Since risk aversion implies $\mathbb{E}[\varepsilon U'(y)] < 0$, the optimum satisfies

$$V > C'(\mu). \tag{21}$$

Interpretation: Risk aversion creates underinvestment relative to first best. The firm lowers the piece-rate intensity to insure the worker, but weaker incentives reduce investment.

4.6 Risk-averse tournament

The worker's expected utility in a symmetric two-person tournament is

$$E[U] = \frac{1}{2}U(W_1 - C(\mu^*)) + \frac{1}{2}U(W_2 - C(\mu^*)). \tag{22}$$

The zero-profit condition is

$$V\mu^* = \frac{W_1 + W_2}{2}. \tag{23}$$

The worker's investment condition can be written as

$$C'(\mu^*) = \frac{2\left[U(W_1 - C(\mu^*)) - U(W_2 - C(\mu^*))\right] g(0)}{U'(W_1 - C(\mu^*)) + U'(W_2 - C(\mu^*))}. \tag{24}$$

Interpretation: The prize spread must compensate for risk and provide incentives. A larger spread increases incentives but makes income riskier.

4.7 Risk aversion and the piece-rate/tournament comparison

A piece rate transforms output linearly, while a tournament transforms a continuous output distribution into a two-point prize distribution.

- Piece rates keep income close to the output distribution, with scaled noise.
- Tournaments truncate extreme outcomes but place probability mass on two prizes.
- The preferred scheme depends on the shape of utility and the amount and type of output noise.

Interpretation: Tournaments can be a crude insurance device in some environments but can be unattractive when the low prize creates too much downside utility loss.

4.8 Common shocks and relative performance evaluation

Suppose measured output contains an individual error and a common activity shock:

$$q_{i\tau} = q_i + \rho_\tau + \nu_{i\tau}. \tag{25}$$

The common shock ρ_τ affects all workers in the same activity. It matters under a piece rate but cancels out in a rank-order comparison.

Interpretation: Rank-order pay can insure workers against common shocks. This is a key advantage of relative performance evaluation: workers are not penalized for bad firm-level or activity-level conditions that affect everyone.

4.9 Heterogeneous workers and adverse selection

Suppose workers differ by ability or cost type. Pure contests among the same type can be efficient. However, when types are private information, low-quality workers may want to enter high-quality contests because high-quality contests have larger expected prizes.

Interpretation: Tournaments create sorting incentives. A compensation system that motivates effort may also attract the wrong contestants if types are not observable.

4.10 Known types and handicapping

If types are common knowledge, the firm can use handicaps. Let h be the handicap given to the lower type, and let $\Delta\mu = \mu_a^* - \mu_b^*$. In a mixed contest, the efficient prize spread can be written as

$$\Delta W = \frac{V}{g(\Delta\mu - h)}. \tag{26}$$

Interpretation: Handicaps neutralize ability differences while preserving marginal incentives. With known types, heterogeneous workers can compete efficiently in the same organization.

5 Identification and theoretical strategy

Because this is a theory paper, identification means theoretical identification: which assumptions generate each result and why the mechanism is isolated.

5.1 Strategy 1: isolate incentives from production complementarities

The model assumes additive production so that worker interactions do not drive the results. This isolates the incentive role of rank-order pay.

Interpretation: If tournaments can implement efficient incentives even in the additive benchmark, their value does not depend on complementarities or team production.

5.2 Strategy 2: use competition to pin down average wages

Free entry and competitive labour markets impose zero expected profit. This separates the average prize level from the prize spread.

- Average prize level: determined by participation and zero profit.
- Prize spread: determined by incentive compatibility.

Interpretation: The paper's key economics is in the spread, not in the average wage.

5.3 Strategy 3: compare piece rates and tournaments under the same technology

Both contracts operate on the same output process and cost function. The comparison therefore focuses on compensation design, not differences in production.

Interpretation: This makes the equivalence result under risk neutrality especially sharp: the same first-best investment can be implemented by two very different institutions.

5.4 Strategy 4: add risk aversion after establishing the benchmark

The paper first proves equivalence under risk neutrality and then introduces risk aversion. This separates incentive efficiency from risk-sharing efficiency.

Interpretation: The risk-aversion section shows that the institution chosen in practice may depend on risk allocation, not only incentive provision.

5.5 Strategy 5: use heterogeneity to expose sorting problems

The paper then allows worker quality to differ. This shows that tournament design is also an assignment problem.

Interpretation: The adverse-selection result warns that competitive tournaments do not automatically produce efficient worker-firm matching when types are private information.

5.6 Main assumptions and what they buy

- **Single period.** Avoids dynamic complications and focuses on career investment incentives.
- **Additive separable production.** Removes team complementarities.
- **Independent idiosyncratic noise.** Makes rank comparisons tractable.
- **Risk-neutral firms.** Firms maximize expected profits and can diversify worker risk.
- **Costly or imperfect monitoring.** Explains why input-based wages are not enough.

Interpretation: These assumptions make the core mechanism transparent. The authors then discuss how sequential contests, sorting, and information revelation could extend the model.

6 Tables and figures

6.1 Table 1: Constant Relative Risk Aversion

Read:

- Table 1 compares optimal piece-rate investment μ, optimal tournament investment μ^*, expected utility under the piece rate $E(U)$, and expected utility under the tournament $E(U^*)$.
- The utility function has constant relative but declining absolute risk aversion.
- When $y_0 = 100$ and $s(y_0) = 0.005$, the tournament is preferred for low values of σ^2 but loses for high values in the table.
- When $y_0 = 25$ and $s(y_0) = 0.020$, the tournament is preferred only for very low variance, and the piece rate dominates as variance rises.
- Investment is generally lower under tournaments than under piece rates in the examples.

Interpretation: The table shows that the risk-aversion comparison is not monotone or universal. Tournaments truncate tails but create a two-prize income distribution. Piece rates preserve the continuous output distribution but expose workers to output noise. Which is better depends on wealth, absolute risk aversion, and noise.

Economic message: Under risk aversion, tournaments are not always better and not always worse. Their value depends on whether the insurance gain from truncating outcomes dominates the utility cost of a low-prize state.

TABLE 1

CONSTANT RELATIVE RISK AVERSION

σ^2	μ	μ^*	$E(U)$	$E(U^*)$
		$y_0 = 100;\ s(y_0) = .005$		
.1	.9995	.9984	5.012155	5.012465
.5	.9975	.9922	5.012150	5.012445
1	.9950	.9846	5.012100	5.012295
3	.9852	.9552	5.011940	5.011925
6	.9710	.9142	5.011800	5.011415
12	.9436	.8420	5.011420	5.010515
		$y_0 = 25;\ s(y_0) = .020$		
.1	.9980	.9938	2.524665	2.524725
.2	.9960	.9878	2.524616	2.524575
1	.9807	.9419	2.524237	2.523437
12	.8094	.5741	2.519930	2.514282

NOTE.—$U = \alpha y^{\alpha}$; $y = y_0 + I + rq - C(\mu)$ for piece rate; $y = y_0 + W_i - C(\mu)$ for contest ($i = 1, 2$); $\alpha = .5, V = 1, C(\mu)$

Constant Relative Risk Aversion

6.2 Figure 1: Income Distribution under Piece Rates and Tournaments

Read:

- Figure 1 shows the continuous income density under a piece-rate scheme and the two discrete prize outcomes under a tournament.
- The piece-rate income distribution has a continuous density over income.
- The tournament converts income into two mass points: the loser prize and the winner prize.
- The figure visually explains why risk aversion changes the comparison between contracts.

Interpretation: The figure emphasizes that a tournament is not a small perturbation of a piece rate. It is a nonlinear transformation of output into income. Workers face a different risk distribution even when expected income and incentives are comparable.

Economic message: The figure clarifies why first-moment equivalence under risk neutrality is not enough under risk aversion. The whole income distribution matters.

6.3 Figure 2: Revenue Functions by Worker Type and the Sorting Problem

Read:

- Figure 2 compares revenue or payoff functions for two worker types, labelled a and b.
- Type a is the higher-quality type and has an efficient investment μ_a^*.
- Type b has a lower efficient investment μ_b^*.
- The $R_b(\mu)$ curve lies southwest of the $R_a(\mu)$ curve.
- The diagram supports the paper's claim that lower-quality workers prefer to enter the higher-quality league rather than self-select efficiently.

Interpretation: The figure captures the adverse-selection problem. Even without production complementarities, the prize structure of high-quality contests attracts low-quality workers. Competitive tournaments alone do not screen types when type is private information.

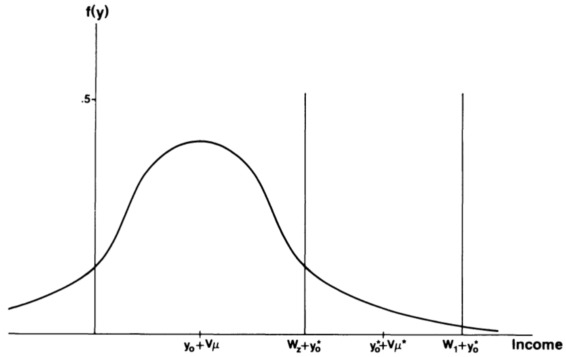

Fig. 1

Figure 1: Income Distribution under Piece Rates and Tournaments

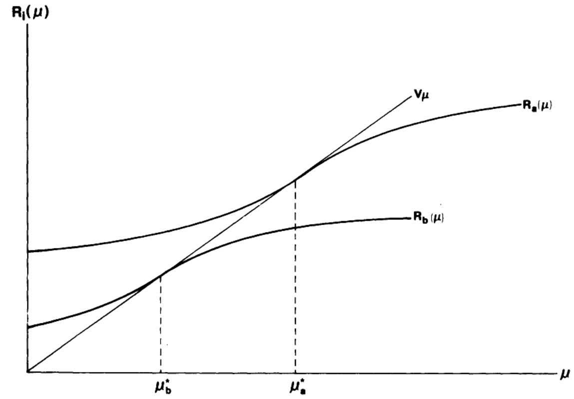

Fig. 2

Figure 2: Revenue Functions by Worker Type and the Sorting Problem

Economic message: The key message is that incentive compatibility within a league is not enough. A full tournament system must also solve participation and sorting across leagues.

7 Interpretive synthesis

7.1 Why promotion prizes can be efficient

The model explains why a large wage jump after promotion can be rational. The wage jump is not a payment for the immediate productivity increase at the promotion date. It is a prize that influences the behaviour of all contestants before the promotion.

7.2 Why rank-order pay reduces measurement demands

Cardinal performance measurement is costly in many occupations. If it is cheaper to rank workers than to quantify their individual marginal products, tournaments economize on measurement costs.

7.3 Why relative performance evaluation can insure common shocks

When all workers face the same firm-level shock, a piece rate exposes workers to that shock, but a tournament does not. This gives tournaments an insurance advantage when common shocks are large.

7.4 Why tournaments can create inefficient sorting

Prize spreads attractive to high-ability workers may also attract low-ability workers. Without observable types or handicaps, low types contaminate high-quality leagues, undermining efficient assignment.

7.5 How the paper connects to later literatures

The model anticipates later work on executive compensation, relative performance evaluation, promotion tournaments, wage inequality within firms, incentives in organizations, career concerns, and sorting in labour markets.

8 Short summary

- The paper models rank-order tournaments as an optimal labour contract.
- A worker chooses costly investment that raises expected output.
- Output contains both effort/investment and random luck.
- Piece rates pay according to output; tournaments pay according to rank.
- Under risk neutrality, both contracts can implement the first-best investment level.
- In tournaments, the prize spread provides incentives; the prize level handles participation.
- The optimal prize spread is inversely related to the density of the performance difference at zero.
- Tournament prizes can be interpreted as a fair gamble over bonds posted by contestants.
- With risk aversion, piece rates and tournaments are no longer equivalent because they generate different income distributions.
- Tournaments can insure common shocks because common shocks cancel in rank comparisons.
- With heterogeneous workers and private information, tournaments can produce adverse selection.
- If worker types are known, handicapping can restore efficient mixed contests.
- The paper explains large promotion prizes, executive wage gaps, and rank-based evaluation as efficient incentive devices.

9 Conclusion

9.1 Core conclusion

Lazear and Rosen show that rank-order tournaments can be optimal labour contracts. When workers are risk neutral, tournaments can induce exactly the same efficient investment as piece rates. The prize spread replaces the marginal piece rate as the incentive instrument.

9.2 Economic intuition

Workers invest more when a marginal increase in output raises the probability of winning a valuable prize. The firm does not need to pay for the level of output; it only needs to set a prize spread that makes workers internalize the social return to investment. This is why ordinal performance information can be enough.

9.3 What the paper shows

The paper shows that wage differences inside firms can be incentive devices, that rank-order compensation can be efficient, that measurement costs matter for contract design, and that heterogeneity introduces important sorting problems.

9.4 What the paper does not show

The model is not an empirical test. It does not estimate prize spreads in actual firms, does not model rich organizational hierarchies, and does not fully solve dynamic promotion systems. It also abstracts from sabotage, collusion, multitasking, and behavioural responses that later tournament models study.

9.5 Final takeaway

The enduring lesson is that wages are not only payments for current output. In organizations, wages can be prizes that shape effort, skill acquisition, sorting, and career incentives. The paper provides the foundational theory for interpreting promotion ladders and executive pay as rank-order tournaments.